

Name: AAVISHKAR SINGH AAVISHKAR SINGH | Reg. No: RA2511003011024

Unit 5 : Session 11 : SLO 1 | Memory-Mapped vs Port-Mapped I/O

I. Multiple Choice

1. B) Memory locations where MMIO treats device registers as memory addresses.
2. B) x86 CISC where port-mapped (isolated) I/O is characteristic of x86.
3. C) IN/OUT where the special instructions used for port-mapped I/O.

II. Fill in the Blanks

1. memory where MMIO shares its address space with main memory.
2. IN/OUT where PMIO uses special instructions such as IN and OUT.
3. RISC where MMIO is commonly used in RISC processors (ARM, MIPS).

III. True or False

1. TRUE where MMIO uses ordinary LOAD/STORE instructions.
2. FALSE where PMIO has a separate port address space.
3. TRUE.

IV. Match the Following

- A) MMIO where 2) Uses memory addresses.
B) PMIO where 1) Uses ports.
C) LOAD where 3) Read from memory.
D) IN where 4) Read from port.

V. Numerical Answers

1. $0x1004$ where base $0x1000$ + offset 4.
2. 50 where the polling loop checks the status 50 times.
3. 256 where an 8-bit port address gives $2^8 = 256$ ports.